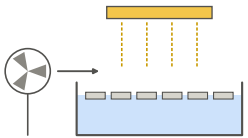
Laboratory

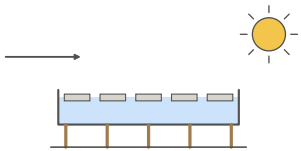
about 0.06–1.5 m^2; fan wind, lamps or no radiation



- r_{open} ↑ — ϕ and gap regime r/δ
- r_{open} ± — energy balance of the elements (lamps or none)
- sets δ — convection: Ra in still air, Ri in weak wind

Pan or tank outdoors

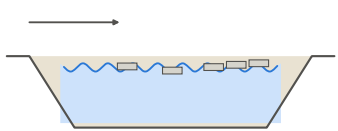
about 0.005–5 m^2, most near 1 m^2; natural forcing



- E_o — sun, wind and humidity of the site
- E_o, E_c — rim, walls and bird screen act on both
- E_o — pan rate is not a lake rate

Farm dam or pond

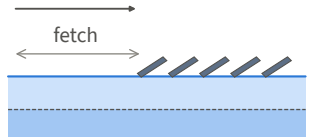
$10–10^4$ m^2, 1–5 m deep; waves, weeks of storage



- r_{under} ↑ — waves wet the elements
- ϕ(t) ↓ — drift, blow-out and clustering of elements
- r_{open} ± — heat storage shifts the saving in season

Reservoir

10^5 m^2 and larger; air adjusts over kilometres



- r_{open} ↑ — fetch: humidified air, lower E_o (gap covers)
- r_{open} ↓ — array shelter and wake: layout, wind rose
- r_{open} ? — r/δ set by array and wind; not measured
- r_{open} ± — stratification: mixed layer sets T_g
- ϕ(t) — level and area change through the year
- S(t) ↓ — ageing: wetting, fouling, wear

Evidence in the synthesis (studies with S and ϕ on the whole surface)

7 studies

94 conditions

measured indoors

30 studies

151 conditions

measured outdoors
+ 1 field pond of pan size

3 studies

23 conditions

measured in the field
S/ϕ 0.54–0.90, all below 1

no measurement

of S with ϕ in full text

6 model studies (37 scenarios)
1 eddy-covariance study, abstract only, ϕ not stated
127 records assume a factor